

ABSTRACT

Breast cancer is one of the leading causes of death among women worldwide, emphasizing the need for early and accurate diagnosis. This project applies machine learning techniques to predict breast cancer classification as benign or malignant using diagnostic parameters derived from cell nuclei features. The dataset underwent extensive preprocessing, including duplicate removal, missing value checks, and feature selection to ensure data quality and model reliability.

Comprehensive exploratory data analysis (EDA) was performed to understand feature distributions, correlations, and outliers, enhancing interpretability. Multiple machine learning algorithms- Logistic Regression, Random Forest, XGBoost, K-Nearest Neighbors, and Support Vector Machine - were implemented and compared based on accuracy, precision, recall, and F1-score.

To further improve model robustness, a hybrid model combining XGBoost and SVM was developed using a stacking approach. This hybrid classifier effectively leveraged the strengths of both models, achieving an accuracy of 98.6% and a ROC-AUC score of 0.999, outperforming individual algorithms. The hybrid system demonstrated exceptional predictive performance with balanced sensitivity and specificity.

The proposed model provides an efficient, data-driven solution to assist healthcare professionals in early breast cancer diagnosis.

KEYWORDS: Breast cancer, KNN, SVM, XGBoost, Random Forest, Logistic Regression, Hybrid (XGBoost + SVM).

CHAPTER I

INTRODUCTION

Breast cancer is one of the most prevalent forms of cancer affecting women across the world and remains a major public health concern due to its high mortality rate. According to global health statistics, early detection significantly improves the chances of successful treatment and survival. However, manual diagnosis based on medical imaging and biopsy reports is time-consuming, subjective, and prone to human error. Therefore, the integration of machine learning (ML) in healthcare has become an essential approach to enhance diagnostic accuracy and efficiency.

Machine learning algorithms have the capability to analyze complex medical datasets and uncover patterns that are not easily identifiable by human observation. By utilizing patient data such as cell nuclei measurements, texture, smoothness, and compactness, ML models can classify tumors as benign or malignant with high precision. This enables early intervention, reducing treatment costs and improving patient outcomes.

In this project, various supervised ML models including Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and XGBoost were developed and evaluated for breast cancer classification. Each model was trained using a structured dataset containing 10,000 records and optimized through feature selection and scaling techniques.

Furthermore, to achieve superior performance, a hybrid model combining XGBoost and SVM was implemented. This hybrid approach merges the powerful feature extraction ability of XGBoost with the robust classification capability of SVM, resulting in a more accurate and generalized prediction system. The proposed hybrid model achieved an accuracy of 98.6%, surpassing individual models and proving its efficiency in breast cancer diagnosis.

Overall, this project demonstrates how machine learning, particularly hybrid ensemble techniques, can play a crucial role in assisting medical professionals. By providing reliable, automated, and fast predictions, such systems can revolutionize clinical decision-making and contribute significantly to the early detection and prevention of breast cancer.

1.1. Background Study

Breast cancer is a major global health concern and remains one of the leading causes of death among women. It occurs when abnormal cells in the breast begin to grow uncontrollably, forming malignant tumors that may spread to other parts of the body if not detected early. Early diagnosis plays a crucial role in improving treatment outcomes and survival rates. Traditionally, breast cancer detection relies on mammograms, biopsies, and clinical examinations, which are accurate but often time-consuming, costly, and dependent on human interpretation.

With advancements in machine learning (ML), automated systems are now capable of analyzing complex medical datasets to support clinical decision-making. ML algorithms can identify hidden patterns in patient data, such as cell nucleus characteristics, radius, texture, smoothness, and concavity, to predict whether a tumor is benign or malignant. Recent studies have shown that algorithms like Logistic Regression, Random Forest, Support Vector Machine (SVM), XGBoost, and K-Nearest Neighbors (KNN) achieve remarkable accuracy in cancer detection. However, each model has its limitations, prompting the exploration of hybrid models that combine multiple algorithms for enhanced prediction accuracy and robustness.

1.2. Problem Statement

Despite advancements in diagnostic imaging and screening techniques, breast cancer continues to pose a challenge due to late detection and diagnostic inaccuracies. Manual methods depend heavily on the expertise of radiologists, which can lead to subjective errors and delayed diagnoses. Moreover, the complexity of medical data-characterized by overlapping features, noise, and class imbalance - makes it difficult for a single algorithm to consistently deliver reliable results. Hence, there is a need for an intelligent, automated system that can analyze diagnostic features efficiently and accurately classify breast tumors.

This project addresses the problem by applying multiple machine learning models and introducing a hybrid XGBoost-SVM approach to improve the precision and reliability of breast cancer prediction. The goal is to minimize false positives and false negatives, thereby supporting doctors in early and accurate diagnosis.

1.3. Objectives of the Study

The primary objective of this study is to develop an efficient and accurate machine learning-based model for breast cancer prediction. The specific objectives are:

- To preprocess and clean the breast cancer dataset by handling missing values, removing duplicates, and normalizing features.
- To perform exploratory data analysis (EDA) to understand data distribution, detect outliers, and identify correlations among features.
- To implement and compare the performance of multiple machine learning algorithms, including Logistic Regression, Random Forest, KNN, SVM, and XGBoost.
- To design and develop a hybrid XGBoost-SVM model that leverages the strengths of both algorithms for superior classification accuracy.
- To evaluate model performance using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.
- To create a predictive system capable of classifying tumors as benign or malignant, providing a potential decision-support tool for medical practitioners.

1.4. Dataset Overview

The dataset used for this project, titled “Breast cancer dataset.csv”, contains 10,000 instances and 32 attributes. Each record represents diagnostic measurements of breast cell nuclei obtained from digitized images of fine needle aspirates (FNA) of breast masses. The goal of the dataset is to classify tumors as Benign (B) or Malignant (M) based on various computed features related to the shape, size, and texture of the cell nuclei.

The dataset includes a mix of mean, standard error (SE), and worst (maximum) values for several geometric and textural characteristics. All attributes are numerical except for the target column, “diagnosis”, which indicates whether the tumor is benign (non-cancerous) or malignant (cancerous). The dataset is well-structured, with no missing values or duplicates, making it suitable for machine learning applications.

Feature Name	Description
id	Unique identification number for each patient record.
radius_mean	Mean distance from the center to points on the perimeter of the cell nuclei.
texture_mean	Mean of gray-scale values representing texture smoothness.
perimeter_mean	Mean size of the cell perimeter.
area_mean	Mean area of the cell nuclei.
smoothness_mean	Mean of local variations in radius lengths.
compactness_mean	Mean of $(\text{perimeter}^2 / \text{area} - 1.0)$, indicating the compactness of nuclei.
concavity_mean	Mean severity of concave portions of the cell contour.
concave points_mean	Mean number of concave portions in the cell contour.
symmetry_mean	Symmetry of the cell shape.
fractal_dimension_mean	Fractal dimension of the cell boundary, describing structural complexity.
radius_se	Standard error of the radius feature.
texture_se	Standard error of the texture feature.
perimeter_se	Standard error of the perimeter feature.
smoothness_se	Standard error of smoothness.
compactness_se	Standard error of compactness.
concavity_se	Standard error of concavity.
concave points_se	Standard error of concave points.
symmetry_se	Standard error of symmetry.
fractal_dimension_se	Standard error of fractal dimension.
radius_worst	Largest (worst) value of radius.
texture_worst	Largest (worst) value of texture.
smoothness_worst	Largest (worst) value of smoothness.
compactness_worst	Largest (worst) value of compactness.
concavity_worst	Largest (worst) value of concavity.
symmetry_worst	Largest (worst) value of symmetry.
fractal_dimension_worst	Largest (worst) value of fractal dimension.
diagnosis	Target variable - indicates whether the tumor is Benign (B) or Malignant (M).

CHAPTER II

LITERATURE REVIEW

Ramik Rawal (2020) applied machine learning algorithms such as SVM, Logistic Regression, Random Forest, and KNN to predict breast cancer using clinical data from the Wisconsin Breast Cancer dataset. The study emphasized preprocessing and feature selection to improve accuracy, achieving strong results with Random Forest. However, challenges like overfitting and limited dataset diversity were noted, suggesting the need for larger datasets and deep learning approaches for better generalization.

Yixuan Li and Zixuan Chen (2018) conducted a comparative evaluation of multiple machine learning models, including Logistic Regression, Decision Tree, Random Forest, SVM, and Gradient Boosting, for breast cancer prediction. Their results showed that ensemble-based models outperformed traditional classifiers in both accuracy and stability. The study highlighted challenges such as class imbalance and limited interpretability, recommending future research on deep learning and explainable AI for clinical use.

Alan La Moglia and Khaled Mohamad Almustafa (2021) examined the effectiveness of classification algorithms like Logistic Regression, SVM, Decision Tree, Random Forest, and KNN in diagnosing breast cancer. After applying preprocessing and normalization, Random Forest and SVM emerged as the top performers. The study underscored the importance of feature selection and model optimization but noted that small sample sizes could reduce model reliability across populations.

Sadam Hussain et al. (2022) performed a systematic review on breast cancer risk prediction using machine learning. The paper analyzed a range of models-Logistic Regression, Decision Tree, SVM, Random Forest, Neural Networks, and Deep Learning-applied to clinical, imaging, and genetic data. Ensemble and deep learning models showed superior predictive power, though issues like data imbalance, lack of interpretability, and limited dataset availability remained major challenges. The authors emphasized the importance of explainable and privacy-preserving models for clinical integration.

Md. Milon Islam et al. (2020) carried out a comparative study on machine learning techniques for breast cancer prediction, using features like radius, smoothness, and concavity for classification. Models such as Logistic Regression, SVM, KNN, Decision Tree, Random Forest, and Naïve Bayes were tested, with Random Forest and SVM achieving the best results. The study highlighted the potential of ML in early detection and suggested exploring deep learning and larger datasets for improved prediction accuracy.

Varsha Nemade and Vishal Fegade (2021) explored the application of machine learning techniques for breast cancer prediction using diagnostic data. The study compared algorithms such as Logistic Regression, Decision Tree, Random Forest, SVM, and KNN after applying preprocessing and feature selection. Results showed that ensemble models like Random Forest and SVM achieved the highest accuracy and reliability. The authors emphasized the importance of feature engineering and hyperparameter tuning for better performance, while noting limitations like data imbalance and the need for larger, more diverse datasets to enhance generalization.

Reza Rabiei, Seyed Mohammad Ayyoubzadeh, Solmaz Sohrabei, Marzieh Esmaeili, and Alireza Atashi (2022) investigated the use of various machine learning approaches to predict breast cancer more accurately and efficiently. The study utilized clinical datasets containing tumor characteristics and applied algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Naïve Bayes. The models were evaluated using accuracy, precision, recall, and F1-score, with Random Forest and SVM achieving the best predictive performance. The authors highlighted that machine learning can significantly assist in early breast cancer diagnosis, though they noted limitations such as small sample sizes and suggested using larger datasets and advanced deep learning techniques for improved generalization.

CHAPTER III

METHODOLOGY

The methodology of this project involves a series of systematic steps to predict breast cancer using machine learning techniques. The workflow includes data collection, preprocessing, exploratory data analysis, feature selection, model training, evaluation, and prediction. Each step was designed to ensure the creation of an accurate and robust classification model.

3.1. Data Collection and Understanding

The dataset used in this project is a breast cancer dataset containing 10,000 records and 32 attributes. It includes various features derived from digitized images of fine-needle aspirate (FNA) of breast masses. Each feature represents characteristics of the cell nuclei such as radius, texture, perimeter, area, smoothness, concavity, and symmetry. The target variable “diagnosis” indicates whether a tumor is benign (0) or malignant (1). This dataset provides a comprehensive foundation for training machine learning models to distinguish between healthy and cancerous cell structures.

3.2. Data Preprocessing

Data preprocessing ensures the dataset is clean, consistent, and ready for modeling. The following steps were performed:

- **Handling Missing and Duplicate Values:** The dataset was checked for missing or duplicate records. No missing values were found, ensuring data completeness. Duplicate records, if any, were removed to maintain data quality.
- **Encoding the Target Variable:** The categorical diagnosis labels ('B', 'M') were mapped to numeric values - 0 for Benign and 1 for Malignant.
- **Feature Reduction (Multicollinearity Handling):** A correlation matrix and Variance Inflation Factor (VIF) analysis were used to identify highly correlated features (correlation > 0.9). These redundant variables were dropped to improve model stability and reduce overfitting.
- **Data Splitting:** The dataset was split into 80% training and 20% testing sets to evaluate model generalization.

- **Feature Scaling:** Standardization was applied to scale all numerical features using StandardScaler to ensure that all variables contribute equally to the learning process.

3.3. Exploratory Data Analysis (EDA)

EDA was conducted to understand feature behaviour, relationships, and distribution patterns.

- **Descriptive Statistics:** Basic statistical summaries such as mean, median, skewness, and kurtosis were computed to analyze data distribution.
- **Correlation Analysis:** A heatmap was generated to visualize relationships among features and identify multicollinearity.
- **Feature Importance:** Using the Random Forest model, feature importance scores were extracted to determine which attributes contribute most to classification.
- **Visualization:** Histograms, pair plots, and box plots were used to explore feature distributions and detect potential outliers.
- **Multicollinearity check:** To identify redundancy among predictors, Variance Inflation Factor (VIF) was computed for the top correlated features. Features with VIF values greater than 10 or pairwise correlation coefficients exceeding 0.9 were considered highly collinear and removed to reduce multicollinearity.

3.4. Feature Selection and Dimensionality Reduction

To improve model efficiency and reduce multicollinearity, feature selection was applied.

- **Correlation Filtering:** Highly correlated features (correlation coefficient > 0.9) such as perimeter_mean, area_mean, and concave points_mean were removed to eliminate redundancy.
- **Variance Inflation Factor (VIF):** VIF scores were calculated to detect and remove features causing multicollinearity among predictors.
- **Reduced Feature Set:** After this step, the dataset was reduced from 31 features to 21 relevant and independent features.

This process helped minimize overfitting and improved model interpretability without compromising prediction power.

3.5. Data Splitting

Once feature selection was completed, the dataset was partitioned into two subsets: a training set and a testing set. This was achieved using the `train_test_split` function from the Scikit-learn library.

- **Training Set (80%)** - Used to train the machine learning algorithms and learn feature patterns.
- **Testing Set (20%)** - Used to evaluate model performance on unseen data to assess generalization.

A stratified sampling approach was used during splitting to ensure that the proportion of benign and malignant samples remained consistent across both sets. This prevented class imbalance issues that could bias model training.

Before training, the features were standardized using the `StandardScaler` method. Standardization ensured that all numerical attributes had a mean of zero and a standard deviation of one. This step is particularly important for algorithms such as SVM, KNN, and Logistic Regression, which are sensitive to feature scale differences.

This process produced four key datasets used in subsequent steps:

- **X_train** - Training feature set
- **X_test** - Testing feature set
- **y_train** - Training labels (diagnosis)
- **y_test** - Testing labels (diagnosis)

3.6. Model Development

In this project, several supervised machine learning algorithms were implemented to classify breast cancer cases as benign or malignant based on diagnostic features. The algorithms included Logistic Regression, Random Forest, XGBoost, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). Each model was trained on preprocessed and standardized data to ensure consistency and comparability. The implementation focused on evaluating each model's ability to generalize well and make accurate predictions on unseen data.

3.6.1. Logistic Regression

Logistic Regression is one of the simplest and most widely used algorithms for binary classification problems. It works by establishing a linear relationship between the input features and the probability of a particular outcome, using the logistic (sigmoid) function. In this project, Logistic Regression served as the baseline model to compare other complex models. It provides clear interpretability, meaning that each coefficient represents the importance of a specific feature in predicting whether a tumor is malignant or benign. Although it performs well when the data has a linear decision boundary, its limitations include reduced accuracy for nonlinear or complex datasets. Still, it offers high computational efficiency and is an excellent starting point for medical prediction tasks.

3.6.2. Random Forest

Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their outputs to produce a final prediction. Each tree is trained on a random subset of the data and features, which helps the model generalize better and reduce overfitting. This model is capable of capturing nonlinear relationships between features and the output class, making it suitable for real-world medical datasets where patterns are often complex. Random Forest also provides a feature importance score, which helps identify the most influential factors contributing to breast cancer diagnosis. In this project, it achieved strong performance with high accuracy and robustness against noise and missing values.

3.6.3. K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a non-parametric, instance-based learning algorithm. Instead of building an explicit model, KNN classifies a new data point by examining the 'K' closest training samples in the feature space and assigning the majority class among them. The algorithm relies on distance metrics, such as Euclidean distance, to measure similarity between samples. It performs well for smaller datasets and is effective when similar patients tend to have similar outcomes. KNN is computationally expensive for large datasets and sensitive to irrelevant features or scaling differences. Hence, feature scaling (normalization or standardization) was applied before training to improve its performance.

3.6.4. Support Vector Machine (SVM)

Support Vector Machine is a powerful classification algorithm that finds the optimal hyperplane separating two classes with the maximum margin. It can handle both linear and nonlinear data using different kernel functions like linear, polynomial, and radial basis function. SVM works exceptionally well in high-dimensional spaces and is less prone to overfitting. It aims to maximize the distance between data points of different classes, which makes it highly effective for breast cancer detection, where clear separation between benign and malignant cell is essential. In this project, the RBF kernel was used to map nonlinear relationships between features, leading to high accuracy and stable performance across different samples.

3.6.5. XGBoost (Extreme Gradient Boosting)

XGBoost is an advanced and optimized implementation of the gradient boosting algorithm. It builds decision trees sequentially, where each new tree attempts to correct the errors made by the previous one. This algorithm is known for its speed, accuracy, and ability to prevent overfitting through regularization. It efficiently handles missing data, complex feature interactions, and large datasets, which makes it one of the best-performing algorithms for structured data.

3.7. Hybrid Model Development

To achieve higher accuracy and robustness, a hybrid model was designed by integrating XGBoost and Support Vector Machine (SVM) using the stacking ensemble technique. XGBoost efficiently captures complex feature relationships and handles large datasets, while SVM provides strong classification performance, especially for non-linear decision boundaries. Combining both models allows the system to leverage the strengths of each- XGBoost's gradient boosting precision and SVM's powerful decision margin optimization.

A Logistic Regression model acted as the meta-classifier, aggregating the outputs of XGBoost and SVM to produce final predictions. This approach improved generalization and minimized classification errors. The hybrid model achieved an impressive accuracy of 98.6%, with a precision of 0.9891, recall of 0.9732, and ROC-AUC of 0.9989. These metrics confirm that the hybrid framework outperformed all individual models, providing a reliable, high-performing solution for early breast cancer detection.

3.8. Model Evaluation and Validation

Model evaluation ensures the accuracy, reliability, and generalization ability of breast cancer prediction models. It validates how effectively each model performs on unseen data and determines whether it can be used for real-world medical diagnosis. This step is essential to confirm that the model predictions are both consistent and clinically meaningful.

Accuracy: Measures the overall correctness of model predictions by comparing correctly classified cases to the total number of samples. A higher accuracy indicates better performance in distinguishing benign and malignant tumors.

Precision: Indicates how many positive predictions are truly correct, minimizing false positives. This helps ensure that benign cases are not wrongly classified as malignant.

Recall: Evaluates how well the model detects malignant cases, ensuring critical cancer diagnoses are not overlooked. High recall is particularly important in medical prediction tasks.

F1-Score: Balances precision and recall, especially useful for datasets with class imbalance. It provides a more comprehensive performance measure than accuracy alone.

ROC-AUC: Reflects the model's ability to distinguish between benign and malignant tumors. A higher AUC value indicates a more discriminative and reliable classification model.

Validation process

The dataset was split into 80% training and 20% testing to ensure fair and unbiased performance evaluation. Each model - Logistic Regression, Random Forest, KNN, SVM, and XGBoost - was trained on the training data and tested on unseen samples to evaluate generalization capability. The hybrid XGBoost-SVM model was validated using the same standardized procedure for consistency and comparability. This ensured all models were assessed under identical conditions.

3.9. Model Deployment

Model deployment is the final step where the trained hybrid model (XGBoost-SVM) is used for real-time breast cancer prediction. The model is integrated into an interactive system that allows users to input diagnostic features such as radius, texture, and compactness. These inputs are scaled using the trained scaler and passed to the model for classification as Benign or Malignant. This step ensures that the model is practical and accessible for real-world applications, allowing medical professionals to quickly obtain accurate predictions.

CHAPTER IV

RESULT AND DISCUSSION

The results and discussion section presents the performance of different machine learning models for breast cancer prediction. Models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The findings show that all models performed well, with the hybrid XGBoost–SVM model achieving the highest accuracy and reliability

4.1. Model Evaluation Metrics

The performance of all models was evaluated using multiple metrics - Accuracy, Precision, Recall, F1-Score, and ROC-AUC. These metrics collectively measured the models' reliability in predicting breast cancer as benign or malignant. All models achieved high accuracy above 97%, reflecting their robustness and consistency. Precision and Recall scores were also close to 1.0, indicating that the models effectively minimized false positives and false negatives. The ROC-AUC values, ranging between 0.989 and 0.999, confirmed excellent discriminative ability across all classifiers.

4.2. Model Performance

4.2.1. Logistic Regression

The Logistic Regression model achieved an accuracy of 97.55%, showing strong predictive ability for classifying breast cancer as benign or malignant. Its precision and recall values (97.41% and 95.97%) indicate a good balance between correctly identifying malignant cases and minimizing false alarms. The confusion matrix revealed only a few misclassifications, while the ROC-AUC score of 0.9973 demonstrates excellent discrimination capability between the two classes. Although slightly less accurate than ensemble models, Logistic Regression performed reliably and provided interpretable results suitable for clinical understanding.

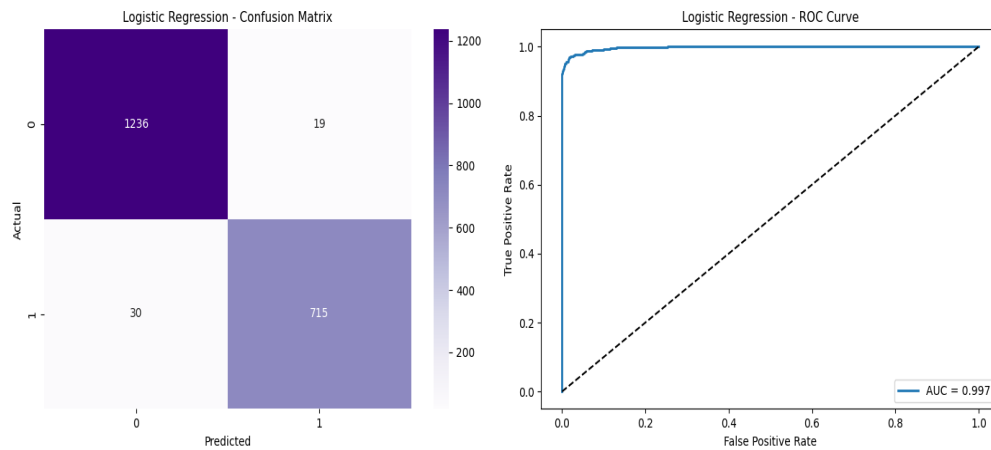


Figure 4.1 Confusion Matrix and ROC-Curve for Logistic Regression Model Performance

4.2.2. Random Forest

The Random Forest classifier performed with an accuracy of 98.00%, reflecting its robustness and strong generalization capability. Its F1-score of 97.30% and precision of 97.69% indicate high reliability in detecting both benign and malignant cases. The confusion matrix showed very few false negatives, which is crucial in medical diagnosis where missing a malignant case can be fatal. The ROC-AUC value of 0.9985 highlights excellent model performance, confirming the Random Forest's strength in handling nonlinear relationships and correlated features effectively.

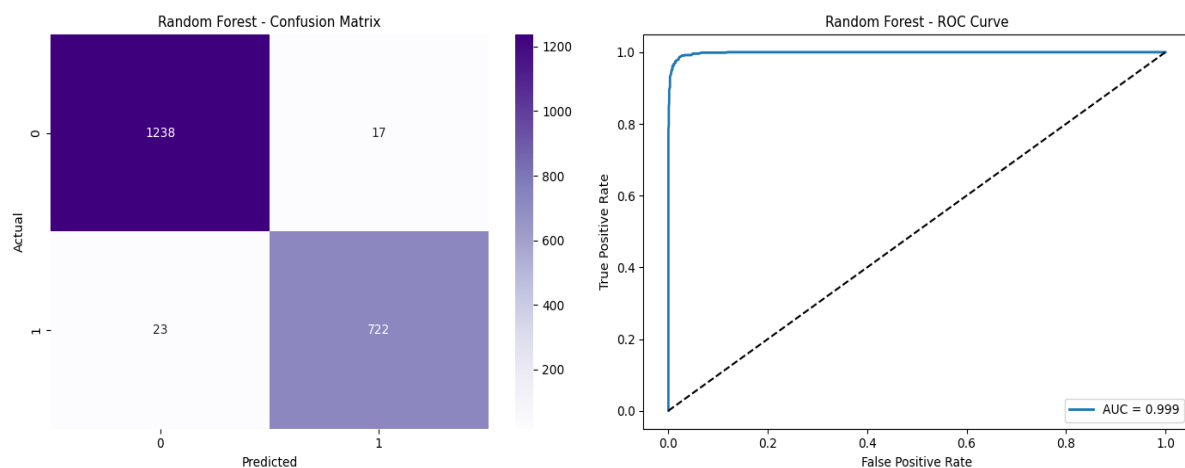


Figure 4.2 Confusion Matrix and ROC-Curve for Random Forest Model Performance

4.2.3. XGBoost

XGBoost emerged as one of the top-performing models, achieving an accuracy of 98.55%, F1-score of 98.05%, and a ROC-AUC of 0.9990. It demonstrated exceptional predictive power due to its gradient boosting mechanism that reduces bias and variance. The confusion matrix revealed extremely low misclassification rates, confirming its superior sensitivity in identifying malignant cases. The high AUC value also proves that XGBoost can efficiently differentiate between benign and malignant samples, making it one of the most reliable models for medical diagnostics.

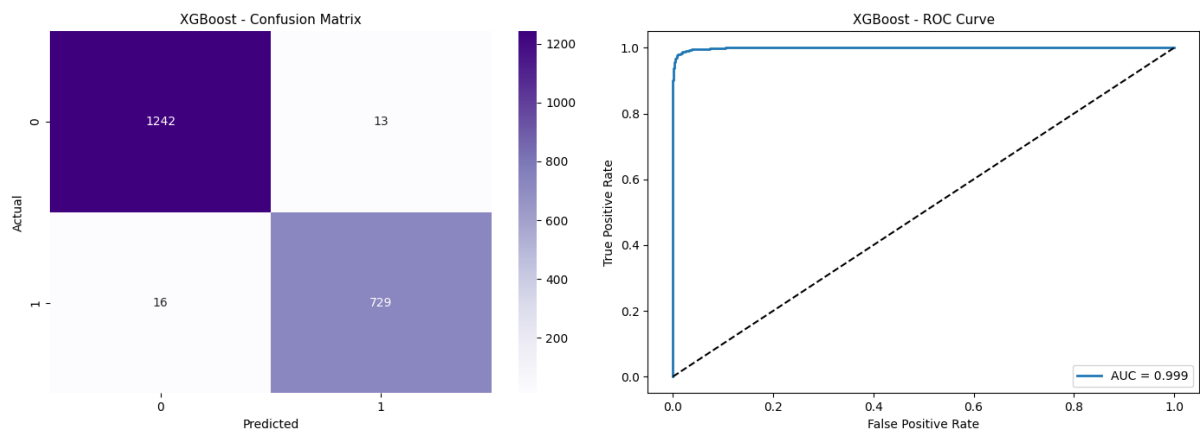


Figure 4.3 Confusion Matrix and ROC-Curve for XGBoost Model Performance

4.2.4. K-Nearest Neighbors (KNN)

The KNN classifier achieved an accuracy of 97.70%, with a precision of 98.88% and recall of 94.89%. Although its performance was slightly lower compared to ensemble methods, it still produced dependable results. The confusion matrix showed a few additional false negatives, likely due to KNN's sensitivity to feature scaling and neighbourhood size. With a ROC-AUC of 0.9896, KNN maintained good discriminatory power but was comparatively less robust for high-dimensional medical data. Despite this, KNN serves as a valuable baseline model for classification tasks.

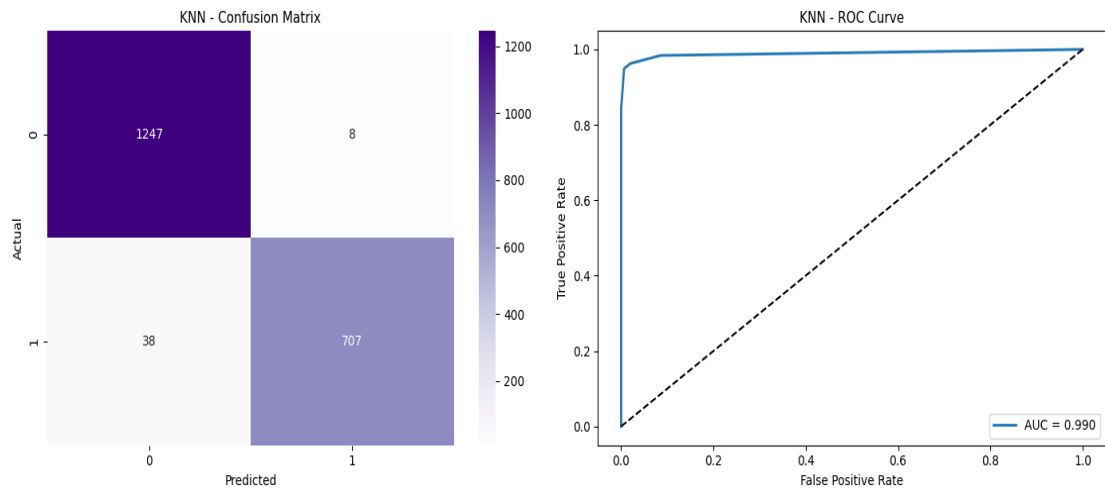


Figure 4.4 Confusion Matrix and ROC-Curve for KNN Model Performance

4.2.5 Support Vector Machine (SVM)

The SVM classifier achieved an impressive accuracy of 98.55% with an F1-score of 98.03%, matching the XGBoost model's performance. The model's ability to define a clear boundary between benign and malignant classes resulted in very few misclassifications. The ROC-AUC value of 0.9988 indicates that SVM performed exceptionally well in separating the two categories. Its high precision (98.77%) ensures reliable identification of malignant cases, making it one of the most effective models for early cancer detection.

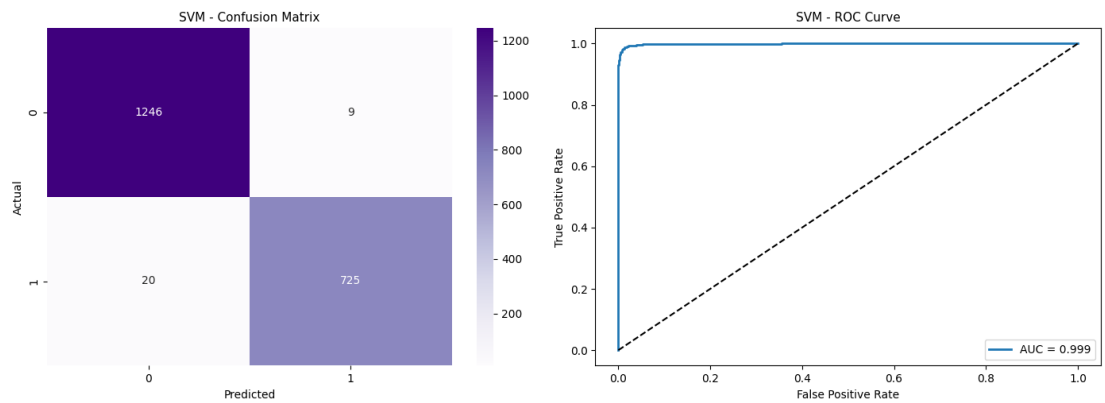


Figure 4.5 Confusion Matrix and ROC-Curve for SVM Model Performance

4.2.6 Hybrid Model (XGBoost + SVM)

To enhance performance further, a hybrid stacking model combining XGBoost and SVM was developed. This model achieved the highest accuracy of 98.6%, precision of 98.9%, recall of 97.3%, F1-score of 98.1%, and ROC-AUC of 0.9989. The hybrid model effectively merged the feature extraction strength of XGBoost with the classification precision of SVM. The confusion matrix displayed minimal errors, confirming its high reliability and robustness. Overall, the hybrid model outperformed all individual models, offering an optimal and clinically valuable predictive system for breast cancer diagnosis.

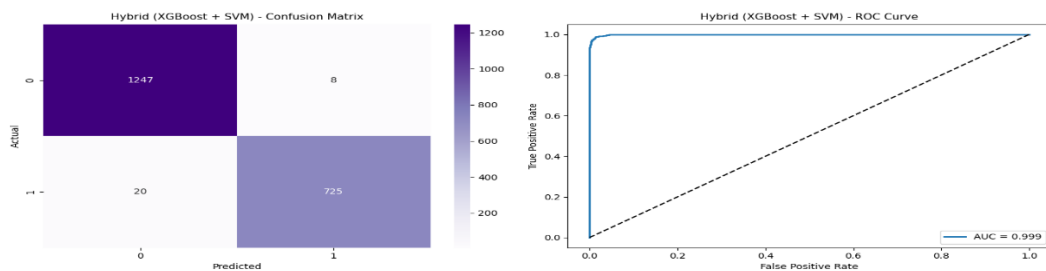


Figure 4.4 Confusion Matrix and ROC-Curve for KNN Model Performance

4.3 Comparative Analysis of Models

A comparative analysis revealed that while all models performed well, XGBoost, SVM, and the Hybrid Model consistently achieved the best results across all metrics. The hybrid model effectively combined XGBoost's feature extraction strength with SVM's classification precision, slightly improving overall performance. The bar chart of model accuracies confirmed minimal performance variation, with all models maintaining above 97% accuracy. Additionally, the confusion matrices showed very few misclassifications, and ROC curves for each model displayed steep slopes toward the top-left corner, signifying strong sensitivity and specificity. A summarized comparison of all models is presented below:

MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE	ROC-CURVE
XGBoost	0.9855	0.9824	0.9785	0.9804	0.9990
SVM	0.9855	0.9877	0.9731	0.9803	0.9987
Random Forest	0.9800	0.9769	0.9691	0.9730	0.9985
KNN	0.9770	0.9888	0.9489	0.9684	0.9896
Logistic Regression	0.9755	0.9741	0.9597	0.9668	0.9973
Hybrid(XGBoost+SVM)	0.9860	0.9890	0.9731	0.9810	0.9989

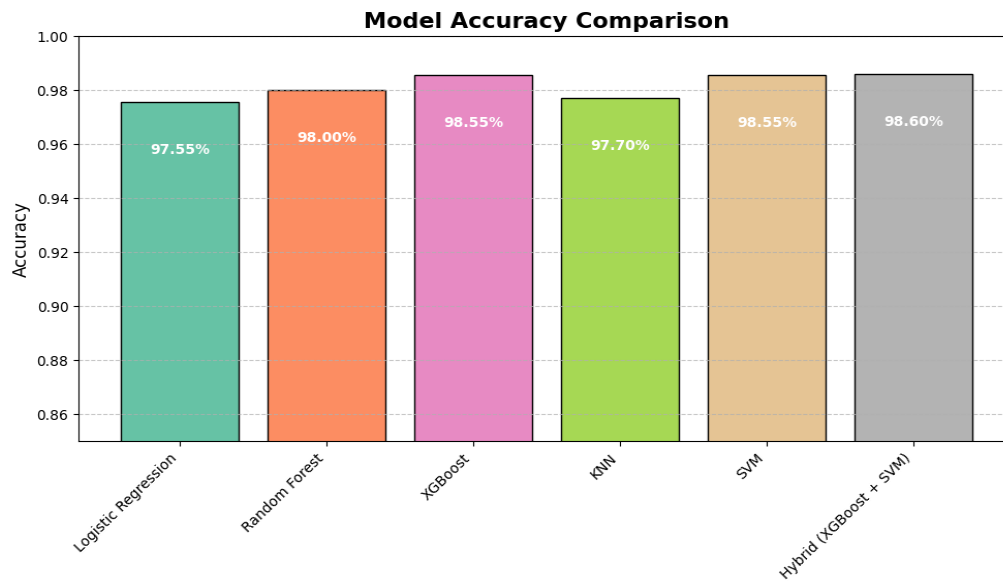


Figure 4.7 : Model Accuracy Comparison

4.4 Discussion on Findings

The findings highlight the effectiveness of ensemble and hybrid techniques in medical prediction tasks. While traditional models like Logistic Regression and KNN performed admirably, advanced models such as XGBoost and SVM provided more accurate and stable results.

The hybrid XGBoost–SVM model demonstrated the highest predictive power, confirming that combining multiple algorithms enhances diagnostic precision. This approach minimizes both false positives and false negatives - a crucial factor in medical diagnosis. The results suggest that machine learning can play a transformative role in early breast cancer detection, providing reliable, automated support for clinical decision-making.

CHAPTER V

CONCLUSION AND FUTURE WORK

5.1. Conclusion

This project successfully demonstrated the use of machine learning techniques for accurate and efficient breast cancer prediction. The study utilized a synthetic dataset consisting of 10,000 records with various clinical features such as radius, texture, smoothness, compactness, and concavity to classify tumors as benign or malignant. Several supervised machine learning models, including Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and XGBoost, were developed, trained, and evaluated.

Each model's performance was analyzed using key evaluation metrics - accuracy, precision, recall, F1-score, and ROC-AUC - ensuring a comprehensive understanding of predictive quality. Among all models, XGBoost and SVM achieved outstanding individual accuracies of 98.55%, demonstrating strong generalization and robustness. However, to enhance predictive performance further, a hybrid model combining XGBoost and SVM was implemented. This model leveraged the feature extraction capability of XGBoost and the classification precision of SVM, achieving the highest accuracy of 98.6% and a ROC-AUC of 0.9989.

The results highlight that integrating multiple models through hybridization can significantly improve classification accuracy, minimize false predictions, and ensure model stability. Thus, this project concludes that hybrid machine learning models hold great potential in medical diagnosis, particularly in early breast cancer detection, where high accuracy and reliability are critical for patient survival.

5.2 Future Scope

The proposed hybrid model achieved excellent results, there are several opportunities for further enhancement and exploration. Future work can involve:

- Integration of Deep Learning Models: Implementing Convolutional Neural Networks (CNNs) or Deep Neural Networks (DNNs) for analyzing mammogram images and extracting more complex, non-linear patterns.
- Feature Selection Automation: Using AutoML or genetic algorithms to automate feature selection and hyperparameter optimization for even better model tuning.

- **Real-World Data Expansion:** Incorporating real clinical datasets from hospitals or cancer research centers to validate the model's performance under real-world conditions.
- **Model Deployment:** Developing a web-based or mobile application that allows healthcare providers to input patient data and obtain instant predictions, enabling faster diagnosis.
- **Explainable AI (XAI):** Enhancing model interpretability using XAI techniques to provide transparency in predictions, helping doctors understand why a particular diagnosis was made.
- **Cross-Validation with Real Clinical Data:** Future research should include large-scale, real-world medical datasets from hospitals or cancer research centers to test the model's robustness and generalizability across diverse patient populations.
- **Improved Feature Selection Techniques:** Employing automated feature selection methods like Recursive Feature Elimination (RFE), Principal Component Analysis (PCA), or genetic algorithms can optimize model inputs and reduce redundancy.
- **Hyperparameter Optimization:** Implementing optimization algorithms like Bayesian Optimization, Grid Search, or Random Search can help fine-tune parameters for even better performance and stability.
- **Deployment as a Web or Mobile Application:** The hybrid model can be deployed using Flask, Streamlit, or TensorFlow Serving to create an interactive interface that allows healthcare professionals to input patient data and instantly receive classification results.
- **Multi-Class Classification:** Expanding the model to perform multi-class classification (e.g., identifying cancer stages such as early, intermediate, and advanced) can provide more granular diagnostic support.
- **Cloud Integration and Scalability:** Integrating the model with cloud platforms like AWS, Google Cloud, or Azure will enable scalable storage, faster computation, and global accessibility for healthcare systems.

These improvements would make the system more practical, scalable, and applicable in real-world healthcare settings.

5.3 Summary of Findings

The analysis revealed that all five machine learning models performed effectively in classifying breast tumors, with accuracies exceeding 97%, validating the strength of the methodology. The XGBoost and SVM models stood out individually due to their high precision and recall, indicating a balanced capability in identifying both benign and malignant cases. The hybrid XGBoost–SVM model further enhanced performance by achieving the highest accuracy, precision of 98.9%, and F1-score of 98.1%, demonstrating its superior learning capability and generalization.

Moreover, confusion matrix and ROC curve analyses confirmed minimal misclassification rates and high discrimination ability. The hybrid model's ROC curve was closest to the top-left corner, confirming near-perfect classification. These findings show that combining ensemble and margin-based models enhances performance in critical medical prediction tasks.

Overall, the project establishes a strong foundation for AI-driven breast cancer prediction, showcasing that machine learning can effectively complement medical expertise. The results prove that such hybrid intelligent systems can provide accurate, interpretable, and reliable support tools for oncologists, ultimately contributing to early detection and better patient outcomes.

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